UNIVERSIDAD NACIONAL
TORIBIO RODRÍGUEZ DE
MENDOZA DE AMAZONAS

Peru, April 20, 2026

Escuela Profesional de Agronomía
Facultad de Ingeniería y Ciencias Agrarias

Flavio Lozano-Isla
Prof. PhD(c)

T.: +51 999997400
E.: flavio.lozano@untrm.edu.pe
E.: flozanoisla@gmail.com

Rafal Marszalek, PhD
Editor-in-Chief
Scientific Reports

Dear Editor,

I hereby enclose the manuscript entitled "Nested Association Mapping (NAM) Reveals Genetic Architecture of Agronomic Traits in Quinoa" for consideration for publication in *Scientific Reports*.

This study presents the first implementation of a nested association mapping (NAM) in quinoa (*Chenopodium quinoa* Willd.), aiming to dissect the genetic architecture of key agronomic and yield-related traits in Andean germplasm. Two NAM populations were developed using contrasting recurrent parents, a modern cultivar (Salcedo INIA) and a traditional landrace (Kancolla), crossed with diverse founder lines from the Peruvian highlands. Multi-environment field trials combined with low-coverage whole-genome resequencing generated a dataset of 6,046 high-quality SNPs, enabling genome-wide association analyses and genomic prediction modeling.

Our results characterized the genetic structure and diversity of the NAM populations, revealing differentiation between Salcedo INIA and Kancolla families. We identified 26 significant SNPs associated with phenological, architectural, and yield traits. Candidate gene analysis identified 68 genes potentially underlying trait variation. Furthermore, genomic prediction analyses demonstrated moderate-to-high predictive ability, with accuracies reaching up to 0.80, highlighting the potential of genomic selection for accelerating breeding in quinoa. The results provide a novel genomic resource and establish a combined analytical approach for NAM, GWAS, and genomic prediction to support crop improvement in quinoa.

We consider this manuscript suitable for publication in *Scientific Reports,* as it aligns with the journal's scope. This work has not been published previously and is not under consideration elsewhere. All authors have reviewed and approved the manuscript and agree to its submission.

Yours sincerely,

Flavio Lozano-Isla
Corresponding author